

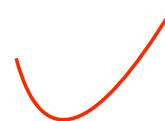
Team Assignment

VG
12/04/2026

Introduction to Machine Components



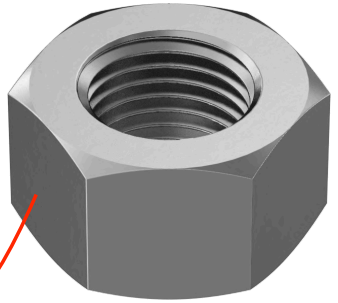
Part No	Name	Qty	Description	Image
1	Gear	1	Gears are versatile mechanical components whose function depends on factors such as their size, shape, number of teeth, and placement within a mechanical system. They are mainly used to transmit power from one component to another. When one gear rotates, it meshes with another gear and causes it to rotate as well. Gears can also be used to modify the torque within a system. For example, when a smaller gear drives a larger gear, the output torque increases, and vice versa. Additionally, gears can be arranged to change the direction of rotation of a shaft when required.	 To what component the gear is transmitting torque and motion to?
2	Internal Retaining Ring	1	Internal retaining rings are small, lightweight metal fasteners installed into a groove machined inside the housing. The ring has an open end that allows it to be compressed during installation and expanded into the groove. In this assembly, the internal retaining ring is used to axially locate and secure the ball bearing within the housing, preventing the bearing from moving along the shaft axis during operation. Once installed, the ring springs back to its original shape and provides a positive mechanical stop, ensuring the bearing remains correctly positioned under load.	



3	Ball Bearing	1	Ball bearings consist of small spherical balls positioned between two concentric rings, known as the inner and outer races. Their primary function is to reduce rotational friction between moving components. They also help prevent wobbling, particularly in applications where precise alignment and smooth rotation are critical, thereby improving overall efficiency. Ball bearings are capable of supporting both radial and axial loads simultaneously. However, due to the relatively small contact areas between the balls and races, they are generally suited for lower load capacities and high-speed (high RPM) applications. Ball bearings are mainly designed to carry radial loads but retain a limited axial load capacity.	 Describe this bearing in relation to Fig 1
4	Housing	1	The housing functions as an enclosure that protects internal components from external contaminants while providing structural support and containment for mechanical elements. It commonly includes mounting features or attachment points, which in this case allow bearings and retaining rings to be securely fixed in position. Housings may also be designed to serve additional purposes such as heat dissipation, sound and vibration damping, ergonomic handling, and the integration of interface ports.	 Describe this bearing in relation to Fig 1
5	Roller Bearing	1	Roller bearings are made up of cylindrical, spherical, or tapered rollers positioned between inner and outer races. Their main function is to minimise rotational friction between moving parts. They also help reduce wobbling in systems where accurate alignment and consistent rotation are essential, improving operational efficiency. Compared to ball bearings, roller bearings typically support higher load capacities for the same size, as they rely on line contact rather than point contact.	

			Roller bearings are designed to withstand radial loads but are generally unable to support axial loads.	
6	Countersunk Screws	6	Countersunk screws are fasteners with a tapered head design that allows the screw to sit flush with the surface after being installed into a countersunk threaded hole. This provides a smooth, level contact between components while minimising surface protrusions. However, as the countersunk hole may not perfectly match the screw head dimensions and depth, complete surface smoothness may not always be achieved. The countersunk screw is used to secure the cap to the housing.	
7	Cap	1	Caps, also referred to as seal housings, are primarily used to secure the roller bearing and rubbing lip seal within the mechanical system. They prevent the bearing and seal from moving axially and becoming dislodged during operation, ensuring stable and reliable performance.	
8	Rubbing Lip Seal	1	Rubbing lip seals provide sealing protection against contaminants such as dirt, dust, corrosive substances, and other foreign particles. They prevent these contaminants from entering the system and causing damage to internal components at the interface between the sleeve, cap, and shaft.	

9	Sleeve (Spacer)	1	Sleeves are installed between the bearing and sprocket and serve as both an intermediary and protective sealing component within the assembly. The sleeve shields the shaft from direct contact with the rubbing lip seal, absorbing wear caused by the seal and preventing damage to the shaft itself. It also allows adjustment of the shaft diameter to ensure proper contact with the rubbing lip seal. Additionally, the sleeve provides the required spacing between the roller bearing and sprocket by acting as a spacer. It further serves as a barrier between the shaft and the external environment, reducing contamination and abrasive damage.	
10	Chain Sprocket	1	Chain sprockets primarily function to transmit power from one shaft to another. When driven by the shaft, the sprocket engages with an external chain connected to another sprocket. As the sprocket rotates, its teeth engage the chain and drive the connected application. This interaction enables the transfer of power between components through the chain system, and vice versa.	
11	Woodruff Key	1	The Woodruff key is fitted into a semi-circular keyseat on the shaft, forming a transition fit between the keyseat and the sprocket to prevent slippage. This allows the sprocket to rotate together with the shaft while being correctly positioned along the axial direction. The transition fit between the Woodruff key, sprocket, and shaft also helps reduce vibration and facilitates easier installation.	

12	Washer	1	Washers placed between the nut and shaft distribute the applied load evenly across the contact surfaces, reducing localised stress concentrations that could otherwise damage components such as the chain sprocket and hexagon nut. They also provide surface protection by acting as a barrier between the nut and the shaft, and increase friction at the interface to achieve a tighter fit, helping to prevent self-loosening and slippage.	
13	Hexagon Nut	1	The hexagon nut is threaded onto the shaft and tightened to form a secure mechanical connection. Together with the washer, it holds the sprocket, roller bearing, and sleeve firmly against the housing, preventing slippage and wobbling of the chain sprocket during shaft rotation.	
14	Profile Key	1	The profile key is inserted into the keyseat to lock the axial position of the shaft relative to the gear. It enables effective power transmission between the shaft and gear while preventing rotational slippage.	
15	External Retaining Ring	1	External retaining rings are small, lightweight metal fasteners that are installed into a groove on the shaft to create a secure stop, preventing axial movement of the gear. They feature a free end that allows the ring to expand during installation. Once installed, the ring springs back to its original shape and grips the shaft firmly. This fastening method is adjustable and allows for easy assembly, disassembly, and maintenance.	
16	Shaft	1	The shaft is a rotating mechanical element, typically with a circular cross-section, used to transmit power or motion from a driving source to an end application. In the assembly shown in Figure 1, the shaft includes two machined keyseats to	

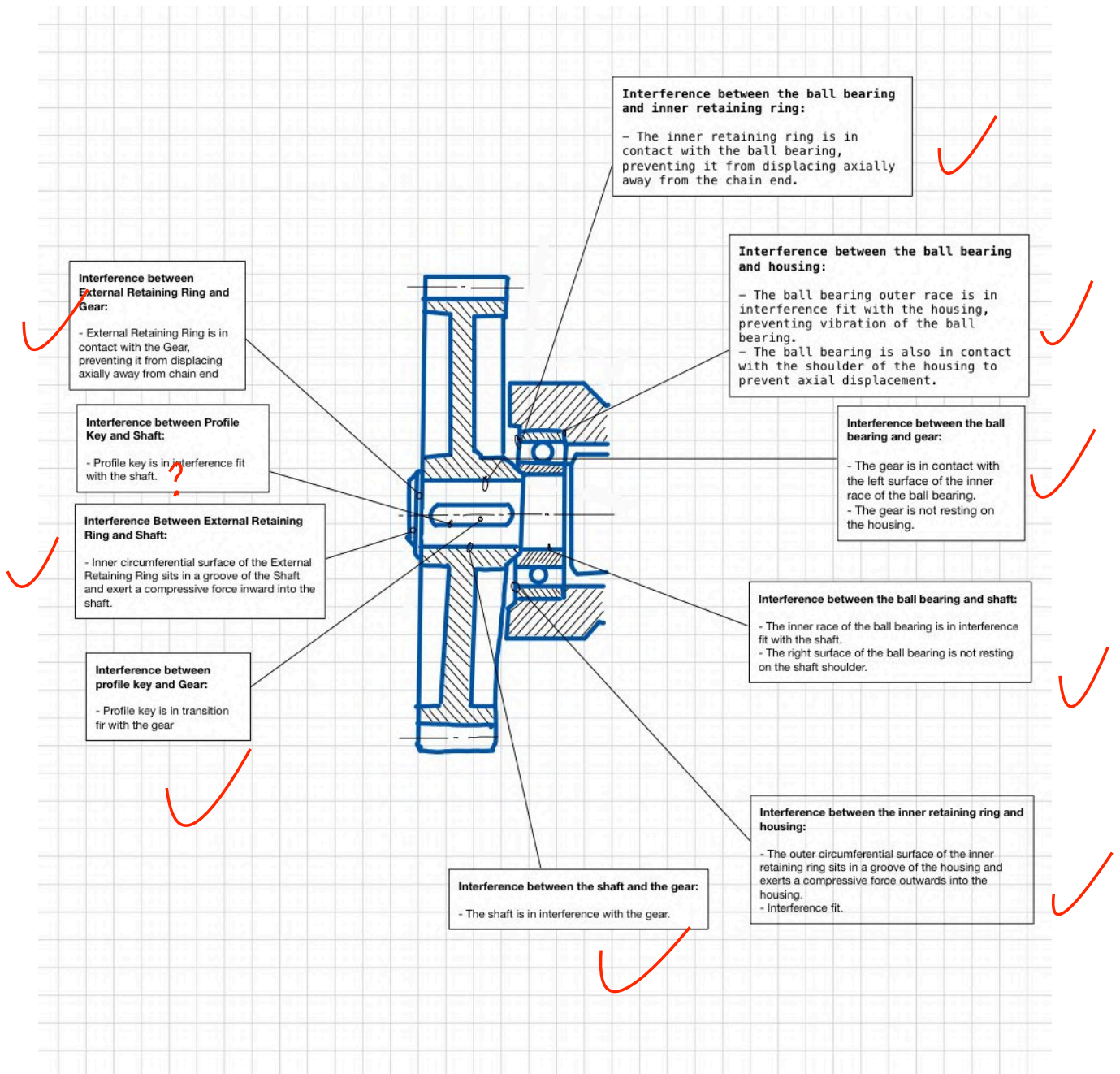
			accommodate the profile key and Woodruff key. It is custom-manufactured with specified shoulders and dimensions to meet the assembly requirements. Additionally, one end of the shaft is threaded to allow tightening of the hexagon nut, while grooves at the opposite end accommodate the external retaining ring.	
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Reference List:

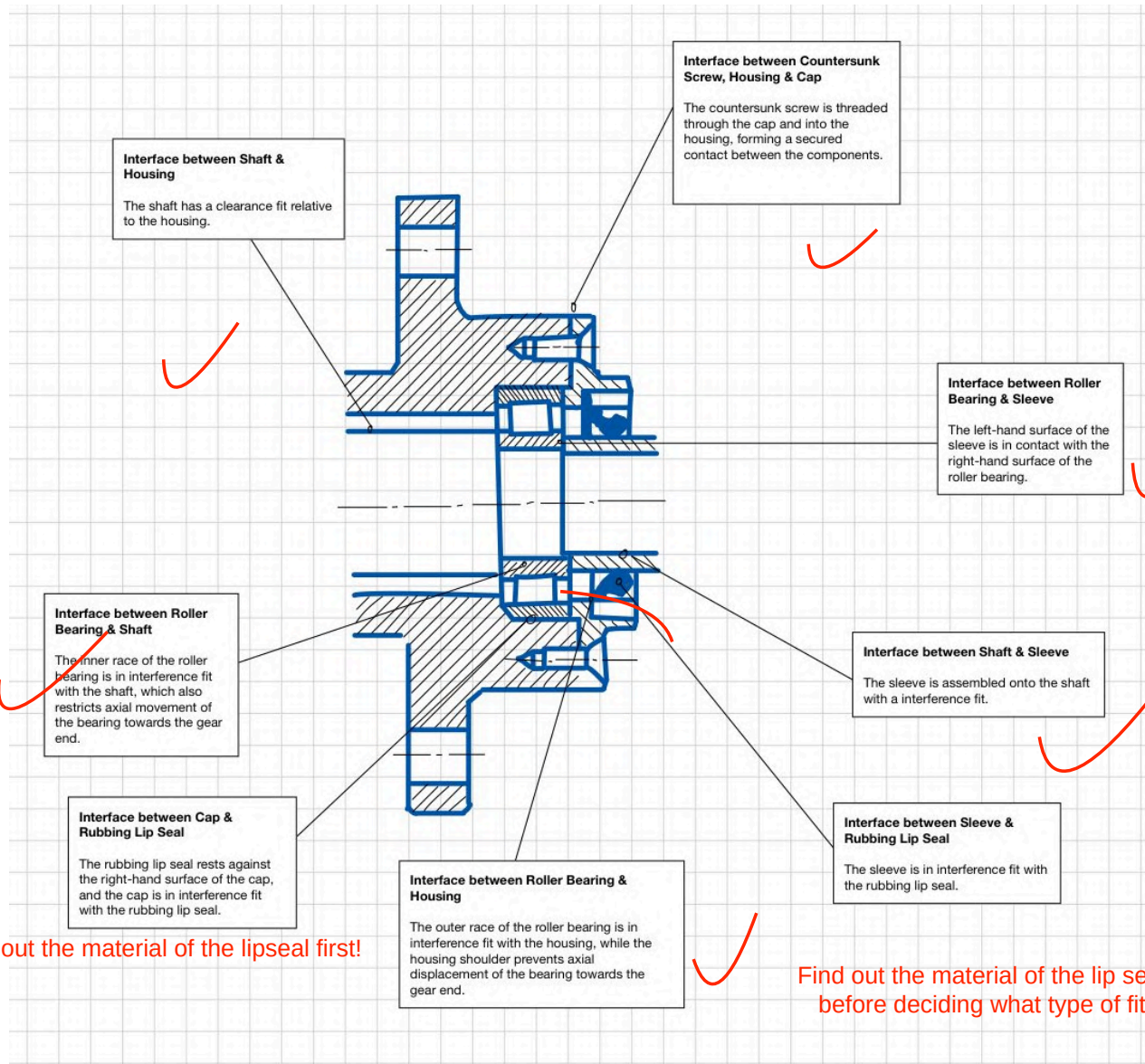
Part	Name	Reference Link
1	Gear	Gear 51T for KLS-1340 kls-04210 – Kent USA Parts
2	Internal Retaining Ring	Rotor Clip HO-90ST PA 5DY33 Retaining Ring, Internal, Circlip, 0.042 Inch Thk, 0.959 Inch Grv Dia., CS Raptor Supplies Singapore
3	Ball Bearing	https://shop.slsbearings.com/ sg/skf-ams-15-abp-angular-c ontact-ball-bearing.html
4	Housing	https://nbsparts.com/lower-be aring-housing-116193/
5	Roller Bearing	https://simmering.pl/en/single -row-tapered-roller-bearings/1 01856-32006-ax-zvl-30x55x1 7-tapered-roller-bearing.html
6	Countersunk Screws	https://www.mcmaster.com/9 0273A830/
7	Cap	https://www.berkeleyjet.com/p -109997-double-drilled-billet- bearing-cap-black-anodized.a spx
8	Rubbing Lip Seal	https://www.grainger.com/pro duct/61CP38?gucid=N:N:FPL :Free:GGL:CSM-1946:tew63 h3:20501231
9	Sleeve (Spacer)	https://www.amazon.sg/20m m%C3%9725mm%C3%9738 -5mm-Alloy-bearing-sleeve-w ear-resistant/dp/B0CZHXN9S 5?th=1
10	Chain Sprocket	https://www.mcmaster.com/2 299K19/
11	Woodruff Key	https://www.grainger.com/pro duct/1AMJ5?gucid=N:N:FPL: Free:GGL:CSM-1946:tew63h

		3:20501231
12	Washer	https://www.walmart.com/ip/1-4-Stainless-Flat-Washer-5-8-Outside-Diameter-100-Pack-Choose-Size-by-Bolt-Dropper-18-8-304-Stainless-Steel/900118907
13	Hexagon Nut	https://www.mcmaster.com/93827A249/
14	Profile Key	https://www.mcmaster.com/90164A807/
15	External Retaining Ring	https://www.mcmaster.com/98398A125/
16	Shaft	https://www.indiamart.com/productdetail/industrial-motor-shaft-23245149548.html

Section 1



Section 2



Section 3

